

Some Rigidity Criteria for Realization Spaces of Phased Matroids

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Roadmap

From matrix minors to matroids

Oriented matroids: universality

Phased matroids: chirotopality and rigidity

Proof idea and sharpness

Conclusion

Minors of a matrix

Maximal minors encode all minors

Let

$$M = [I_r \mid S] \in \mathbb{F}^{r \times n}$$

be in standard form. Every minor of S is a maximal minor of M :

$$\det M[X, Y] = \det M_{[r]X \rightarrow Y}.$$

Example

Consider

$$M = \begin{bmatrix} 1 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & -1 & 3 \end{bmatrix}.$$

$$\det M_{(1,4,5)} = \det \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 1 \\ 0 & -1 & 3 \end{bmatrix} = \det \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}.$$

Combinatorial abstraction: Matroids

For a full-rank matrix M , define

$$\mathcal{B}(M) = \left\{ B \in \binom{[n]}{r} : \det M_B \neq 0 \right\}.$$

- $\mathcal{B}(M)$ satisfies the *basis exchange property*.
- Whitney (1935): abstract this property to define matroids.

Matroid (Whitney, 1935)

A *matroid* is a set system $M = (E, \mathcal{B})$ such that

$$(B1) \quad \emptyset \neq \mathcal{B} \subseteq \binom{E}{r},$$

$$(B2) \quad \forall A, B \in \mathcal{B}, \forall a \in A \setminus B, \exists b \in B \setminus A \text{ such that } A - a + b \in \mathcal{B}.$$

Grassmannian and Plücker coordinates

The Grassmannian consists of all r -dimensional subspaces of $\mathbb{F}^n$.

Each point is represented by a full-rank matrix, modulo row operations:

$$\mathrm{Gr}(r, \mathbb{F}^n) = \mathrm{GL}(r, \mathbb{F}) \backslash \{M \in \mathbb{F}^{r \times n} : \mathrm{rank}(M) = r\}.$$

Each point of the Grassmannian is represented projectively (up to scale) by its maximal minors,

$$\mathrm{Gr}(r, \mathbb{F}^n) \hookrightarrow \mathbf{P}\left(\mathbb{F}^{\binom{[n]}{r}}\right), \quad \mathbb{V}(M) \mapsto [p_I := \det(M_I)]_{I \in \binom{[n]}{r}}$$

called the *Plücker coordinates*.

Grassmann–Plücker relations

These coordinates are not independent. They satisfy homogeneous quadratic relations, whose common zero locus is the Grassmannian.

$$p_{123}p_{145} - p_{124}p_{135} + p_{125}p_{134} = 0.$$

Forget the values, keep only the nonzero pattern

The underlying matroid is determined by the nonzero pattern of the Plücker coordinates.

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Chirotopes

When $\mathbb{F} = \mathbb{R}$, the signs of maximal minors give a refinement of the underlying matroid.

Induced chirotope

A full-rank matrix $M \in \mathbb{R}^{r \times n}$ induces a rank r chirotope on $E = [n]$

$$\chi_M(x_1, \dots, x_r) := \operatorname{sgn} \det(M_{x_1, \dots, x_r}) \in \{0, \pm 1\}.$$

Abstractly, a chirotope is a determinant-like sign pattern: nonzero, alternating, and satisfying sign Grassmann–Plücker relations.

Forget the magnitudes, keep only the sign pattern.

Beyond the zero/nonzero pattern, over $\mathbb{R}$ we can also record the **signs of (maximal) minors**:

$$M = \begin{bmatrix} 1 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & -1 & 3 \end{bmatrix}, \quad \chi_M(1, 4, 5) = +1 = \operatorname{sgn} \det \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}.$$

Oriented matroids and realization space

Oriented matroid

An oriented matroid is given by a chirotope $\chi : E^r \rightarrow \{0, \pm 1\}$, modulo global sign:

$$\mathcal{O} = (E, \{\chi, -\chi\}).$$

Oriented matroid realization space

A point $\mathbb{V} \in \text{Gr}(r, \mathbb{R}^n)$ determines an oriented matroid by the sign pattern of its Plücker coordinates:

$$[p_I(\mathbb{V})]_I \mapsto \mathcal{O}(\mathbb{V}) = ([n], \{\text{sgn}(p_\bullet), -\text{sgn}(p_\bullet)\}).$$

Its realization space is the fiber in the real Grassmannian:

$$\text{Gr}_{\mathbb{R}}(\mathcal{O}) = \{\mathbb{V} \in \text{Gr}(r, \mathbb{R}^n) : \mathcal{O}(\mathbb{V}) = \mathcal{O}\}.$$

Mnëv's universality

However, the geometry of oriented-matroid realization spaces is far from simple, as shown by the following Mnëv's universality theorem.

Mnëv's universality [Mnë91]

For every primary basic semialgebraic set S over $\mathbb{Z}$, there exists a rank-3 oriented matroid $\mathcal{O}$ such that

$$\mathrm{Gr}_{\mathbb{R}}(\mathcal{O}) \text{ is stably equivalent to } S.$$

Murphy's Law in Algebraic Geometry

*“Even moduli spaces of well-behaved objects should be arbitrarily bad unless there is an **a priori reason** otherwise.”*

— Vakil [Vak06]

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Phirotopes and phased matroids

Over $\mathbb{C}$, Below, Krummeck, and Richter–Gebert introduced phirotopes as a complex analogue of chirotopes.

Induced phirotope and phased matroid

Similar to the real setting, a full-rank matrix $M \in \mathbb{C}^{r \times n}$ induces

$$\varphi_M(x_1, \dots, x_r) := \text{ph det}(M_{x_1, \dots, x_r}) = \text{ph } p_{\mathbf{x}}(M) \in \mathbb{S}^1 \cup \{0\},$$

where

$$\text{ph}(0) = 0, \quad \text{ph}(r\alpha) = \alpha \quad (r > 0, \alpha \in \mathbb{S}^1).$$

A phased matroid is obtained from a phirotope modulo global phase:

$$\mathcal{M} = (E, [\varphi]), \quad [\varphi] = \{\alpha\varphi : \alpha \in \mathbb{S}^1\}.$$

Phased matroids and torus action

Phased-matroid realization space

Its realization space is the fiber in the complex Grassmannian:

$$\mathrm{Gr}_{\mathbb{C}}(\mathcal{M}) := \{\mathbb{V} \in \mathrm{Gr}(r, \mathbb{C}^n) \mid \mathcal{M}(\mathbb{V}) = \mathcal{M}\}.$$

Two kinds of column rescaling

The column torus splits as $(\mathbb{C}^\times)^E = \mathbb{R}_{>0}^E \times (\mathbb{S}^1)^E$.

- Positive rescaling preserves the phased matroid:

$$\lambda \in \mathbb{R}_{>0}^E \implies \mathcal{M}(MD(\lambda)) = \mathcal{M}(M).$$

- Rephasing changes the phase labels, but preserves the realization space up to isomorphism:

$$\beta \in (\mathbb{S}^1)^E \implies \mathcal{M}(MD(\beta)) =: \mathcal{M}(M)^\beta, \quad \mathrm{Gr}_{\mathbb{C}}(\mathcal{M}) \cong \mathrm{Gr}_{\mathbb{C}}(\mathcal{M}^\beta).$$

Chirotopality: "essential oriented"

Below, Krummeck, and Richter–Gebert first drew the line, in rank two via cross ratios.

Chirotopality (higher-rank) [Rui13]

A phirotope φ , or its phased matroid $\mathcal{M}$, is called **chirotopal** if it can be **rephased to a chirotope**:

$$\exists \alpha \in \mathbb{S}^1, \beta : E \rightarrow \mathbb{S}^1 \quad \text{such that} \quad \alpha \cdot \varphi^\beta(E^r) \subseteq \{0, \pm 1\}.$$

In this case,

$$\text{Gr}_{\mathbb{C}}(\mathcal{M}(\varphi)) \cong \text{Gr}_{\mathbb{C}}(\mathcal{M}(\varphi^\beta)) \cong \text{Gr}_{\mathbb{R}}(\mathcal{O}(\alpha\varphi^\beta)).$$

e.g. Non-chirotopal:

$$M = \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & -1 & 2i \end{pmatrix}, \quad \text{cr}((1, 2), (3, 4)) = (-1) \cdot (-1)^{-1} \cdot (2i) \cdot (2)^{-1} \notin \mathbb{R}$$

Rigidity Problem and Our Criterion

Rigidity: unique realization up to $GL(r, \mathbb{C})$ and columnwise positive scaling.

Known results and our criterion

Assumption	Result
non-chirotopal, rank two	rigid [BKR03]
non-chirotopal, uniform (every r -subset is a basis)	rigid [Rui13; Sch17]
non-chirotopal, with a $U_{r,r+2}$ -restriction	rigid (this work)

$$U_{r,r+2}\text{-restriction} \iff \exists E' \subseteq E, |E'| = r + 2, \text{ uniform on } E'.$$

The bound $r + 2$ is sharp.

This criterion unifies and extends previous rigidity results.

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Proof idea of Ruiz's uniform criterion

Write $z = a\alpha$, where $a > 0$ and $\alpha \in \mathbb{S}^1$. A dot denotes an unknown magnitude.

Law of sines and base block

LOS:

$a\alpha + b\beta + c\gamma = 0$, $\alpha \not\parallel \beta \implies [a : b : c]$ is determined by α, β, γ .

$$Q = \begin{bmatrix} 1 & b\beta \\ 1 & \dot{a}\alpha \end{bmatrix}, \quad \alpha \not\parallel \beta, \quad \dot{a}\alpha - b\beta = \dot{c} \cdot \text{ph}(\det Q).$$

Thus, $\dot{a}$ is uniquely determined.

Reduction of Ruiz's framework

If the first $r + 1$ columns form a uniform restriction, then WLOG normalize

$$M = [I_r \mid \mathbf{1}_r \mid N].$$

It remains to show that N is determined up to positive column rescaling.

Completing Ruiz's argument

Remaining argument

Uniformity gives $N_{ij} \neq 0$, while non-chirotopality gives some $z \in N \setminus \mathbb{R}$. After positive column rescaling, we may assume

$$M = \left[\begin{array}{c|ccc} & 1 & \cdots & 1 \\ & \vdots & \color{red}{z} & * \\ I_r & 1 & * & * \end{array} \right], \quad \begin{bmatrix} 1 & 1 \\ * & * \end{bmatrix}, \quad \begin{bmatrix} 1 & * \\ 1 & * \end{bmatrix}.$$

The first row of N is normalized. Applying LOS to the two types of base blocks on the right determines the remaining entries.

Beyond the uniform case

Our proof adapts Ruiz's idea to address three new issues:

1. We need more structure around the nonreal entry z .
2. Non-uniformity requires more types of base blocks.
3. What does a $U_{r,r+2}$ -restriction give us?

1. A non-rephasing-real column

A non-rephasing-real column

Non-chirotopality forces a column of S containing two nonparallel phases:

$$[I_r \mid S], \quad S_j \sim \begin{bmatrix} a\alpha \\ \vdots \\ b\beta \end{bmatrix}, \quad \alpha \not\parallel \beta.$$

Column decomposition

We separate the nonbasis columns as

$$M = [I_r \mid \mathbf{1}_r \mid Z \mid R],$$

Z : NRR columns, R : rephasing-real columns.

2. More solvable blocks

Type I / II / III blocks

$$\text{Type I: } \begin{bmatrix} 1 & 1 \\ 1 & a\alpha \\ 1 & \dot{d} \end{bmatrix}, \quad \alpha \not\parallel 1.$$

$$\text{Type II: } \begin{bmatrix} 1 & p & 1 \\ 1 & a\alpha & 0 \\ 1 & p' & \dot{d} \end{bmatrix}, \quad \alpha \not\parallel 1, \quad p \neq p'.$$

$$\text{Type III: } \begin{bmatrix} 1 & p & 1 & 0 \\ 1 & p' & a\alpha & 0 \\ 1 & q & 0 & 1 \\ 1 & q' & 0 & \dot{d} \end{bmatrix}, \quad \alpha \not\parallel 1, \quad p \neq p', \quad q \neq q'.$$

Each block propagates known magnitudes to the unknown $\dot{d}$.

3. What does $r + 2$ give us?

What does a $U_{r,r+2}$ -restriction give us?

Beyond $U_{r,r+1}$, we gain one more column $\mathbf{q}$:

$$M = [I_r \mid \mathbf{1}_r \mid \mathbf{q} \mid \cdots].$$

Its entries are nonzero by the 1×1 minors, and pairwise distinct by the 2×2 minors with $\mathbf{1}_r$.

Sketch of the main theorem

- **Layer 1: solve the NRR anchors.**

The column $\mathbf{1}_r$ and base/Type I blocks determine Z .

- **Layer 2: propagate to the RR columns.**

The extra column $\mathbf{q}$ provides the nondegeneracy required by Type II/III blocks.

(Type III handles the case where $\mathbf{q}$ is rephasing-real.)

Sharpness examples

All M_t induce the same non-chirotopal phased matroid, but they are not equivalent.

$$M_t = \begin{bmatrix} 1 & 0 & 0 & 1 & \textcolor{blue}{1} & 1 \\ 0 & 1 & 0 & 1 & i & 0 \\ 0 & 0 & 1 & 1 & \textcolor{blue}{1} & t \end{bmatrix}, \quad t > 1.$$

(Using the same idea, we also construct sharpness examples in all ranks.)

Remark

Our sharpness examples also show that a previously claimed $r + 1$ -criterion in Schaar's thesis cannot hold in general.

Simultaneously recovers and extends previous criteria

First, non-chirotopality forces $|E| \geq r + 2$.

- **Ruiz–Schaar:** uniform $\implies$ the existence of a $U_{r,r+2}$ -restriction.
- **Below–Krummeck–Richter–Gebert:** in rank 2, non-chirotopality forces the presence of a $U_{2,4}$ -restriction.

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Limitations and Future Directions

1. A matroidal explanation?

Can deletion and contraction explain how local solvability propagates to realization rigidity? Ideally,

solvable blocks + suitable uniform minor $\implies$ solvability.

On the other hand, our counterexamples arise by breaking the solvability of these blocks. (**forbidden matroid minors?**)

2. Beyond matrix algebra?

Our proof recovers magnitudes from local phase relations. Is there an intrinsic language for these relations?

3. Topology at the $r + 1$ boundary

$U_{r,r+1}$ -restriction $\not\Rightarrow$ rigidity.

Is the realization space contractible? Can its dimension be read from combinatorial data?

Thank you